

# Project Design Phase-II

## Technology Stack (Architecture & Stack)

|                |                                                |
|----------------|------------------------------------------------|
| Date:          | 04 October 2026                                |
| Team ID:       | SWTID-2026-5601                                |
| Project Name:  | Auto Ticket Classification using Flow Designer |
| Maximum Marks: | 4 Marks                                        |

### 1. Technical Architecture Overview

The Auto Ticket Classification System leverages ServiceNow's cloud infrastructure, Flow Designer automation engine, and Natural Language Understanding (NLU) / Predict Intelligence components. The architecture is organized across three primary tiers: Client Interface, Platform Orchestration Logic, and Cloud Data/Storage Engine.

- **1. User Request Ingestion:** End-users (Students & Teachers) log incidents via the Service Portal UI or REST API integration.
- **2. NLU & Text Parsing:** ServiceNow NLU engine parses the short description, extracts intent entity keywords, and computes confidence scores for categories.
- **3. Flow Designer Trigger:** Flow Designer executes predefined workflow rules to evaluate conditions, set Category/Subcategory fields, and calculate initial priority.
- **4. Data Store Query & Queue Assignment:** The platform references Assignment Group tables to automatically assign the ticket to the designated IT Support Group.
- **5. Notification Dispatch:** Notification Engine sends automated alerts (Email/SMS) to callers and assigns technicians instantly.

### 2. Table 1: Components & Technologies

| S.No | Component                           | Description                                                                         | Technology                                                   |
|------|-------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------|
| 1    | User Interface                      | Portal interface for students & teachers to submit tickets and track status.        | ServiceNow Service Portal, HTML5, CSS3, AngularJS, Bootstrap |
| 2    | Application Logic-1 (Workflow)      | Automated orchestration, trigger handling, and routing workflow.                    | ServiceNow Flow Designer, JavaScript (ES6 / GlideScript)     |
| 3    | Application Logic-2 (Text Mining)   | Intent classification, keyword parsing, and confidence scoring.                     | ServiceNow Natural Language Understanding (NLU) Engine       |
| 4    | Application Logic-3 (Notifications) | Automated email notifications and caller status dispatch logic.                     | ServiceNow Notification Engine, Notification Triggers        |
| 5    | Database                            | Relational database storing incident records, user profiles, and assignment groups. | ServiceNow Relational Database (MariaDB enterprise cluster)  |

| S.No | Component                       | Description                                                                                         | Technology                                              |
|------|---------------------------------|-----------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| 6    | Cloud Database                  | Cloud-hosted database service managing multitenant data storage.                                    | ServiceNow High Availability Cloud Data Center Storage  |
| 7    | File Storage                    | Storage for ticket attachments (log files, screenshots, diagnostic error reports).                  | ServiceNow Cloud Attachment Storage (AES-256 Encrypted) |
| 8    | External API-1                  | REST API for integrating third-party IT tools and monitoring alerts.                                | REST API / JSON Web Services                            |
| 9    | External API-2                  | User identity authentication API against School Active Directory/LDAP.                              | OAuth 2.0 / Single Sign-On (SSO) / Azure AD Integration |
| 10   | Machine Learning Model          | Supervised classification model for predicting ticket categories based on historic resolution data. | Predictive Intelligence / Text Categorization Model     |
| 11   | Infrastructure (Server / Cloud) | Cloud deployment infrastructure ensuring high availability and fault tolerance.                     | ServiceNow Multi-Instance Enterprise Cloud Architecture |

### 3. Table 2: Application Characteristics

| S.No | Characteristics          | Description                                                                                                  | Technology                                                                                |
|------|--------------------------|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| 1    | Open-Source Frameworks   | Open web libraries and front-end frameworks used in user interface development.                              | AngularJS, Bootstrap 4, jQuery, FontAwesome                                               |
| 2    | Security Implementations | Comprehensive role-based access control, encrypted communications, and OAuth authentication.                 | TLS 1.3, AES-256 Encryption, Role-Based Access Control (RBAC), OAuth 2.0, OWASP Standards |
| 3    | Scalable Architecture    | Multi-tier microservices-enabled platform architecture designed to scale seamlessly with ticket volume.      | ServiceNow Multi-Instance Cloud Architecture & Load Balanced Application Tier             |
| 4    | Availability             | High-availability clustering with automatic failover across redundant data centers.                          | Active-Active Multi-datacenter Redundancy & Load Balancing                                |
| 5    | Performance              | Sub-second response time for classification workflows, supported by caching and asynchronous flow execution. | GlideRecord Query Caching, Asynchronous Event Processing, CDN Assets                      |